

1. Factors that affect antigen and antibody tests (3):

- **Sample quality and handling:** The condition and handling of the sample can affect test results. For example, contaminated or degraded samples can lead to false negatives or false positives.
- **Test sensitivity and specificity:** The test's ability to detect the antigen or antibody accurately can impact results. A test with low sensitivity may miss detecting the antigen, while a test with low specificity may detect non-specific substances.
- **Interfering substances or cross-reactivity:** Substances in the sample can interfere with the test or react with the test reagents, leading to inaccurate results.

2. Antibody formation (5):

- **Antigen presentation to B cells:** Antigens are processed and presented to B cells, which recognize them through their surface-bound antibodies.
- **Activation and proliferation of B cells:** Activated B cells multiply and differentiate into plasma cells, which produce antibodies.
- **Differentiation into plasma cells:** B cells mature into plasma cells, which specialize in antibody production.
- **Class switching and affinity maturation:** B cells can switch to producing different antibody classes (e.g., IgG, IgA) and undergo affinity maturation to produce high-affinity antibodies.
- **Antibody production and secretion:** Plasma cells produce and secrete antibodies, which enter the bloodstream and other bodily fluids.

3. MHC and its classes (5):

- **Major Histocompatibility Complex (MHC):** A group of genes that play a critical role in the immune response.
- **MHC Class I:** Presents intracellular antigens to CD8+ T cells (cytotoxic T cells), which kill infected cells.
- **MHC Class II:** Presents extracellular antigens to CD4+ T cells (helper T cells), which coordinate immune responses.
- **MHC Class III:** Involved in the complement system and inflammation.

- **MHC genes are highly polymorphic:** Many variations exist, allowing for greater diversity in immune responses.

4. Usage of monoclonal antibodies (5):

- **Cancer treatment (e.g., rituximab):** Targets cancer cells for destruction.
- **Autoimmune disease treatment (e.g., adalimumab):** Blocks inflammatory pathways.
- **Infectious disease treatment (e.g., palivizumab):** Neutralizes pathogens.
- **Diagnostic tests (e.g., flow cytometry):** Identifies and characterizes cells.
- **Research applications (e.g., immunohistochemistry):** Studies protein expression and localization.

5. Lymphocyte recirculation (3):

- **Lymphocytes continuously circulate between lymphoid organs and blood:** Allows for immune surveillance and rapid response.
- **Regulated by adhesion molecules and chemokines:** Controls lymphocyte migration and localization.

6. Classes of antibodies:

- **IgA:** Found in mucosal surfaces, provides local immunity.
- **IgD:** Expressed on mature B cells, activates antibody production.
- **IgE:** Involved in allergic reactions and parasite immunity.
- **IgG:** Most common antibody class, provides long-term immunity.
- **IgM:** First response to infections, activates complement system.

7. Physical barriers of innate immune system:

- **Skin and mucous membranes:** Prevents pathogen entry.
- **Respiratory tract cilia and mucus:** Traps and removes pathogens.
- **Gastrointestinal tract acidity and enzymes:** Kills pathogens.
- **Blood-brain barrier:** Restricts pathogen entry into the central nervous system.
- **Eye and ear protective mechanisms:** Prevents pathogen entry.

8. Anaphylaxis:

Anaphylaxis is the form of immune response in which exaggerated response is generated against Allergen. The molecules which produced such response are called as anaphylatoxins.

- Severe, systemic allergic reaction.
- IgE-mediated release of histamine and other mediators.
- Symptoms include hives, itching, swelling, and anaphylactic shock.

9. Affinity maturation:

- Process of increasing antibody affinity for antigens.
- Occurs through somatic hypermutation and selection.
- Results in higher antibody efficacy and specificity.

10. Characteristics of antibodies:

- **Specificity for antigens:** Recognizes specific epitopes.
- **Affinity for antigens:** Strength of antibody-antigen binding.
- **Valence (number of binding sites):** Determines antibody functionality.
- **Isotype (class) and subclass:** Defines antibody function and localization.
- **Idiotypic (unique epitope recognition):** Determines antibody specificity.

11 : Heptan definition:

Heptan is a hydrocarbon molecule with seven carbon atoms (C₇H₁₆). It is a volatile organic compound and a solvent.

12 : Immunology and antigen difference:

Immunology: Study of the immune system, including its functions, disorders, and responses to pathogens and antigens.

Antigen: A substance that triggers an immune response, recognized by immune cells (e.g., proteins, peptides, carbohydrates).

13.What is secondary lymphoid organs.

Secondary lymphoid organs are lymphoid tissues where immune responses are initiated and amplified. Examples include:

- Lymph nodes • Spleen • Peyer's patches • Tonsils • Lymphoid follicles

14 .ELISA Advantages and differentiate between direct and indirect ELISA

1. ELISA (Enzyme-Linked Immunosorbent Assay):

Advantages:

- High sensitivity and specificity
- Easy to perform and automate

- Can detect small amounts of antigen or antibody
- Non-invasive

Direct ELISA:

* Detects antigen directly * Antibody is conjugated to an enzyme * Antigen is immobilized on a plate

Indirect ELISA:

- * Detects antibody against an antigen* Antigen is immobilized on a plate
- * Secondary antibody (conjugated to an enzyme) binds to primary antibody

15.Function of granulocytes:

Granulocytes (neutrophils, eosinophils, basophils) are white blood cells that play a crucial role in the immune response:

- Phagocytosis and elimination of pathogens
- Release of granules containing enzymes and chemicals
- Involvement in inflammation and allergic reactions

16 . T-cell receptor (TCR):

The TCR is a complex of proteins on the surface of T cells that recognizes and binds to antigens presented by major histocompatibility complex (MHC) molecules on antigen-presenting cells (APCs). This recognition triggers T cell activation and initiation of immune responses.

17 .Differentiate between BCR and TCR.

BCR (B Cell Receptor) and TCR (T Cell Receptor) are both crucial components of the immune system, but they have distinct differences:

BCR (B Cell Receptor):

- Found on the surface of B cells
- Recognizes and binds to antigens directly
- Triggers B cell activation, proliferation, and antibody production
- Composed of membrane-bound immunoglobulins (Ig)
- Responsible for humoral immunity

TCR (T Cell Receptor):

- Found on the surface of T cells

- Recognizes and binds to antigenic peptides presented by MHC molecules on APCs
- Triggers T cell activation, proliferation, and initiation of cell-mediated immunity
- Composed of TCR alpha and beta chains (or gamma and delta chains)
- Responsible for cell-mediated immunity

18. Role of Adjuvants in Polyclonal Antibodies Production

Adjuvants are the substances which potentiates immune response against antigens.

For the production of polyclonal antibodies following Adjuvants are used

1. Freund's Complete Adjuvants
2. Freund's Incomplete Adjuvants

Freund's Complete Adjuvants (FCA)

Freund's complete adjuvant (FCA) are water-in-oil emulsion containing antigens with heat killed Mycobacterium tuberculosis. These adjuvants have the ability to stimulate both humoral & cell Mediated immune response for antibody production.

Freund's Incomplete Adjuvants

These adjuvants are also water-in-oil emulsion containing antigens without heat killed Mycobacterium tuberculosis. These are used as booster antigen dose. These are used by mixing Equal parts of the antigens.

19 .Lymphocytes and its function?

Lymphocytes are a type of white blood cell that plays a crucial role in the immune system. There are two main types of lymphocytes:

1. B cells (B lymphocytes):

- Produce antibodies to recognize and bind to specific antigens
- Activate humoral immunity •Provide long-term immunity

3. T cells (T lymphocytes):

- Recognize and kill infected cells or foreign substances
- Activate cell-mediated immunity •Regulate immune responses

Functions of Lymphocytes:

1. Recognition and elimination of pathogens
2. Production of antibodies (B cells)
3. Activation of immune responses (T cells)
4. Regulation of immune responses (T cells)
5. Development of immunological memory (B cells and T cells)

6. Surveillance and monitoring of the body for infections and cancer (T cells)

20 .Phagocytosis is a process in which cells engulf and digest foreign particles, microorganisms, and cellular debris. It is an essential mechanism for:

1. Eliminating pathogens (bacteria, viruses, fungi) 2. Removing dead cells and debris
3. Recycling cellular components 4. Initiating immune responses

Phagocytosis involves:

1. Recognition of particles or microorganisms 2. Extension of pseudopodia (cellular extensions) 3. Engulfment of particles or microorganisms 4. Formation of phagosomes (membrane-bound vesicles) 5. Fusion with lysosomes (containing digestive enzymes)
6. Digestion and degradation of engulfed material

Phagocytic cells include:

1. Neutrophils 2. Macrophages 3. Dendritic cells 4. Monocytes

Phagocytosis plays a crucial role in protecting the body against infection and disease, and dysfunction in phagocytosis can lead to various immune disorders.

21 .Name the cells which are involved in immune system

Cells involved in the immune system:

1. T cells (CD4+ and CD8+) 2. B cells 3. Macrophages 4. Dendritic cells
5. Neutrophils 6. Eosinophils 7. Basophils 8. Natural killer cells

22 .Difference btw T dependent and independent cells

T-dependent cells (TD cells):

- Require T cell help for activation • - Recognize protein antigens • Induce immune responses

T-independent cells (TI cells):

- Do not require T cell help • - Recognize non-protein antigens (e.g., polysaccharides)
- Induce limited immune responses

23 . Briefly describe the monoclonal antibodies usage

Monoclonal antibodies usage:

- Cancer treatment (targeting specific cancer cells) • Autoimmune disease treatment (blocking inflammatory pathways)• Infectious disease treatment (neutralizing pathogens)
- Diagnostic tests (identifying specific antigens)•Research applications (studying protein function and expression)

24 .Explain basic structure of antibodies

- Consist of two heavy chains and two light chains • Have a variable region (antigen-binding site) and constant region • Classes: IgA, IgD, IgE, IgG, IgM

25 .Describe the types of two adjuvants

Types of adjuvants:

1. Freund's adjuvant (complete and incomplete):

- Enhances immune responses
- Induces inflammation and activation of immune cells

2. Alum adjuvant:

- Enhances immune responses • Induces antibody production and T cell activation

26 What is megakaryocytes and granulocytes with their functions

Megakaryocytes:

- Large bone marrow cells •Produce platelets (blood clotting)

Granulocytes:

- White blood cells with granules in cytoplasm • Include neutrophils, eosinophils, and basophils • Involved in phagocytosis, inflammation, and immune responses